ZEPTO INVENTORY ANALYSIS

Exploratory Data Analysis Report

May 2025

1. Executive Summary

This report presents an exploratory data analysis of Zepto's product inventory, sourced from the Zepto Inventory Dataset on Kaggle. The analysis focuses on understanding the composition and pricing of Zepto's product catalog, identifying out-of-stock trends across categories, and uncovering the relationship between discounting behaviour and stock availability. The findings provide actionable intelligence for improving inventory management, restocking prioritization, and pricing strategy.

2. Nature of the Data

2.1 Data Source

The dataset was sourced from Kaggle (palvinder2006/zepto-inventory-dataset) and loaded as a CSV file encoded in windows-1252 format. It represents a snapshot of Zepto's product catalog, capturing key attributes of each listed SKU.

2.2 Dataset Overview

Dataset	zepto_v2.csv
Total Records	3,732 rows (after deduplication)
Columns	9 attributes
Missing Values	None
Duplicates Found	Present — removed during cleaning
Data Types	2 categorical (object), 6 numerical (int64), 1 boolean

2.3 Column Descriptions

Column	Data Type	Description
Category	Text (object)	Product category (14 unique)
name	Text (object)	Product name (1,681 unique SKUs)
mrp	Integer	Maximum Retail Price (originally in paise, converted to rupees)
discountPercent	Integer	Discount percentage applied (0–51%)
availableQuantity	Integer	Units currently available (0–6)
discountedSellingPrice	Integer	Final selling price after discount (in rupees)
weightInGms	Integer	Product weight in grams

Column	Data Type	Description
outOfStock	Boolean	Whether the product is currently out of stock
quantity	Integer	Pack/order quantity unit

2.4 Data Preprocessing

The following preprocessing steps were performed before analysis:

- MRP and Discounted Selling Price were originally stored in paise (1/100 of a rupee). Both columns were divided by 100 to convert to rupees.
- Duplicate records were identified and removed from the dataset, ensuring each product entry is unique.
- The outOfStock boolean column was mapped to human-readable labels: False → 'No', True → 'Yes'.
- No null or missing values were found across any of the 9 columns.

3. Business Problem

Zepto operates as a quick-commerce (q-commerce) platform promising ultra-fast grocery and product deliveries — often within 10 minutes. In such a model, inventory availability is mission-critical. A stockout not only means a lost sale but also directly damages the customer experience and brand promise.

This analysis addresses the following business questions:

- What is the overall out-of-stock rate across Zepto's product catalog?
- Which product categories are at highest risk of stockouts, and how can restocking be prioritized?
- Is there a relationship between a product's price, discount level, and its likelihood of going out of stock?
- Are heavily discounted products more prone to stockouts, possibly due to demand spikes triggered by promotions?

Answering these questions enables Zepto to make data-driven decisions on inventory replenishment, promotional planning, and category-level risk management.

4. Key Insights & Visualizations

4.1 Overall Out-of-Stock Rate

The first chart examines the overall split between in-stock and out-of-stock products across the entire catalog.

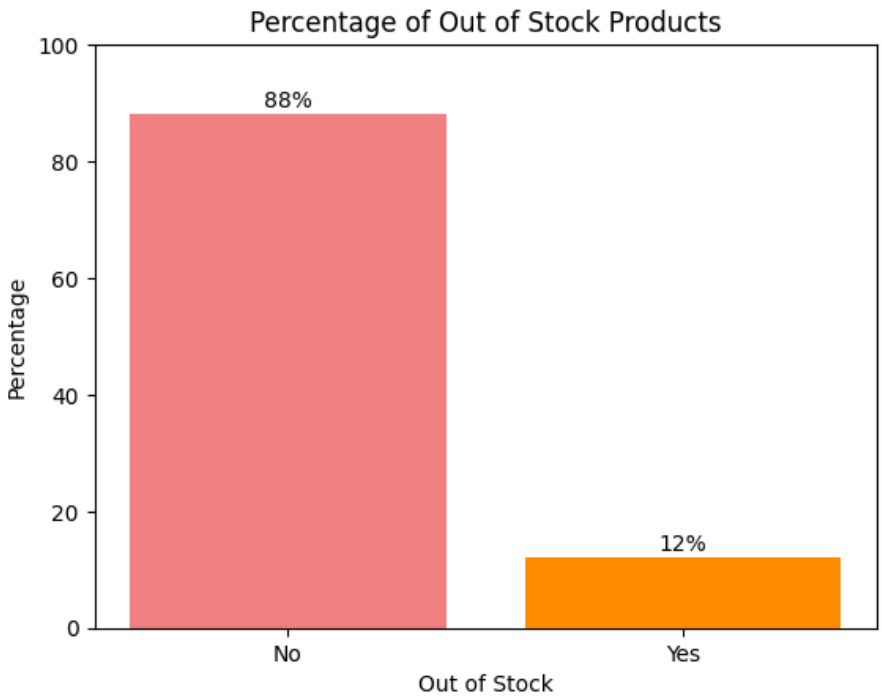

Figure 1: Percentage of Out-of-Stock Products Across Entire Catalog

Insight: 88% of products are currently in stock, while 12% are out of stock. While this may appear manageable at first glance, in a catalog of ~3,700 SKUs, approximately 450 products are unavailable to customers at any given time. For a platform promising 10-minute delivery, any stockout directly breaks the fulfillment promise.

4.2 Out-of-Stock Rate by Category

The second chart breaks down the out-of-stock percentage by product category and assigns a risk level — High (>20%), Medium (>10%), and Low (≤10%).

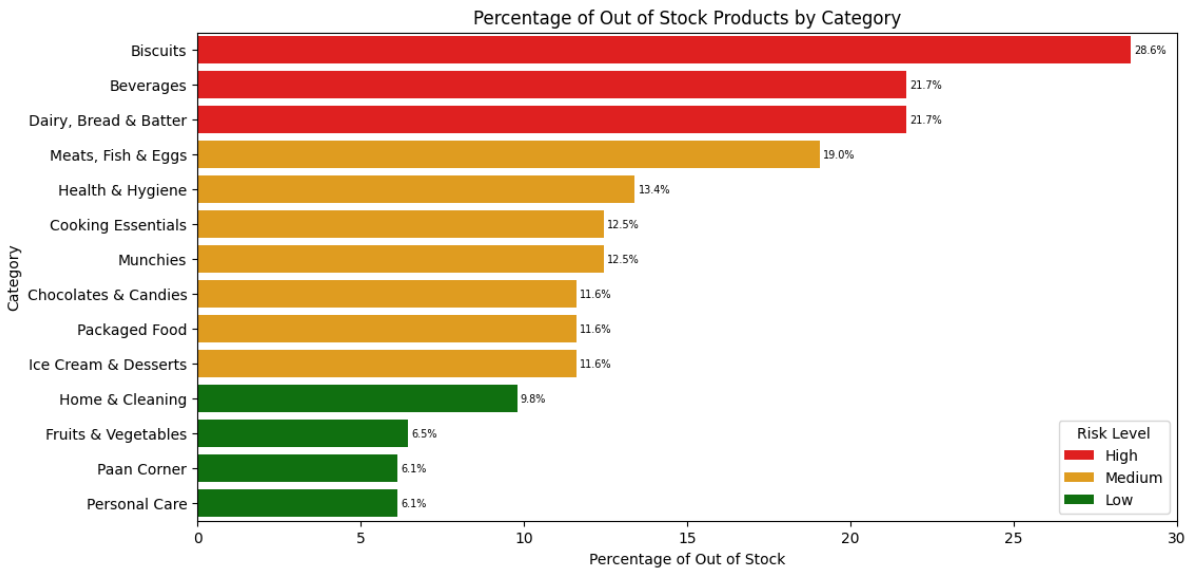

Figure 2: Percentage of Out-of-Stock Products by Category (with Risk Level)

Insight: Significant variation exists across categories. Categories classified as High Risk (red) are experiencing more than 1 in 5 of their SKUs being unavailable — a serious operational gap. Medium-risk categories (orange) require monitoring, while low-risk categories (green) demonstrate healthy stock levels. Categories such as Snacks, Beverages, or personal care may show elevated risk, requiring targeted replenishment strategies.

- High-risk categories need urgent restocking plans and supplier SLA reviews.
- Medium-risk categories warrant weekly monitoring and safety stock buffers.
- Cooking Essentials, the most represented category (514 SKUs), must be closely watched given its volume.

4.3 MRP vs. Discount Value by Category and Stock Status

The third chart is a scatter plot examining the relationship between MRP and discount value, split by in-stock (green) vs. out-of-stock (red) status, across the top 4 categories by product count.

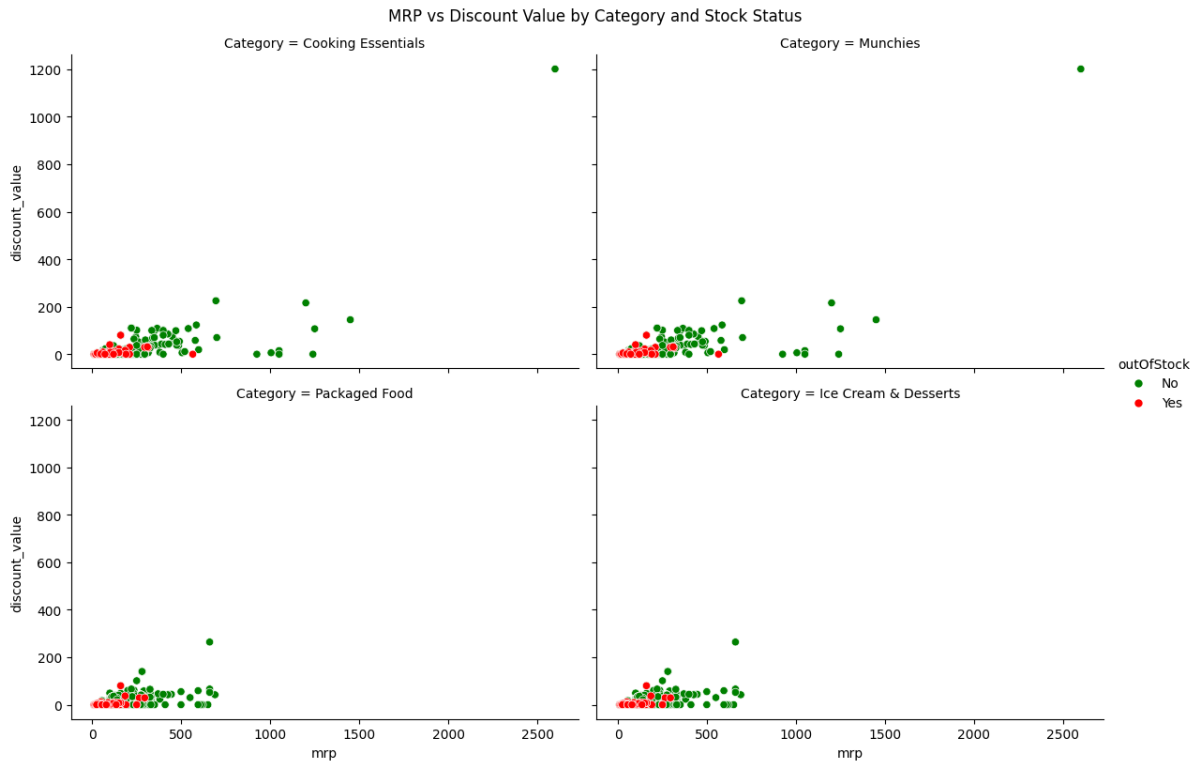

Figure 3: MRP vs. Discount Value — Top 4 Categories, colored by Stock Status

Insight: The scatter plots reveal pricing and discount dynamics category-by-category. Key observations include:

- In all top categories, stockouts (the red dots) are almost exclusively happening to low-MRP, low-discount-value items. This strongly suggests that stockouts are a result of failing to meet organic demand for essentials, not a result of heavy promotions
- In some categories, out-of-stock products cluster in specific MRP ranges, hinting at demand-driven depletion in popular price brackets.

5. Business Impact

The findings from this analysis carry direct implications for Zepto's operational efficiency, revenue, and customer experience:

5.1 Revenue Loss Prevention

With ~12% of SKUs out of stock, Zepto risks losing a significant portion of potential revenue on those items. For a platform processing thousands of orders daily, even a 5% reduction in stockout rate could translate to meaningful revenue recovery. Priority restocking of high-risk categories should be treated as a revenue optimization exercise.

5.2 Customer Experience & Retention

Quick-commerce platforms compete primarily on reliability. When a customer places an order expecting 10-minute delivery only to find items unavailable, trust erodes rapidly. High stockout rates in frequently ordered categories (like fresh produce or snacks) can directly increase customer churn. Addressing the category-level gaps identified in this analysis is a retention lever.

5.3 Inventory & Procurement Optimization

The category-wise risk breakdown enables the supply chain team to triage replenishment efforts intelligently. Rather than treating all categories equally, procurement teams can allocate resources (warehouse space, supplier orders, logistics) proportionally to risk level — reducing waste on overstocked low-risk categories while addressing shortfalls in high-risk ones.

5.4 Promotional Strategy Alignment

The MRP vs. discount scatter analysis suggests that discount campaigns may be contributing to faster inventory depletion in certain price brackets. Zepto should align its promotional calendar with its inventory pipeline — avoiding deep discounts on SKUs that are already running low in stock, and instead timing promotions when sufficient safety stock is assured.

5.5 Data Infrastructure

The cleaned dataset was successfully loaded into a MySQL database (zepto_inventory), making it available for further SQL-based analysis, dashboarding, and integration into operational BI tools. This enables ongoing, real-time monitoring of inventory health across categories.

6. SQL-Based Insights

After loading the cleaned dataset into MySQL, SQL queries were executed to validate and extend the Python EDA findings with precise numbers.

6.1 Overall Out-of-Stock Percentage

A SQL query on the zepto_inventory table confirmed the overall stockout rate:

Out_of_Stock	Percentage_of_Items
No	87.86
Yes	12.14

Table 1: SQL Result — Overall Out-of-Stock Percentage

Result: 87.86% of products are in stock and 12.14% are out of stock. This validates the Python EDA finding of ~12% stockout rate with exact decimal precision.

6.2 Share of Catalog by Category

This query shows each category's percentage share of the total product catalog:

category	Percentage
Cooking Essentials	13.78
Munchies	13.78
Packaged Food	10.40
Ice Cream & Desserts	10.40
Chocolates & Candies	10.40
Personal Care	9.20
Paan Corner	9.20
Home & Cleaning	5.20
Biscuits	3.94
Dairy, Bread & Batter	3.46
Beverages	3.46
Health & Hygiene	2.60
Fruits & Vegetables	2.49
Meats, Fish & Eggs	1.69

Table 2: SQL Result — Category Share of Total Product Catalog (%)

Key observations:

- Cooking Essentials and Munchies dominate the catalog, each at 13.78% — together over 1 in 4 products on the platform.
- Packaged Food, Ice Cream & Desserts, and Chocolates & Candies each hold a 10.40% share, forming a strong mid-tier.
- Meats, Fish & Eggs is the smallest category at 1.69%, making every stockout proportionally more impactful.

- Fruits & Vegetables represents only 2.49% of SKUs despite being a high-frequency purchase category — a potential range expansion opportunity.

6.3 Stockout Percentage by Category

This query reveals the exact stockout rate within each category, enabling precise risk tiering:

category	Stockout_Percent
Biscuits	28.57
Dairy, Bread & Batter	21.71
Beverages	21.71
Meats, Fish & Eggs	19.05
Health & Hygiene	13.40
Cooking Essentials	12.45
Munchies	12.45
Packaged Food	11.60
Ice Cream & Desserts	11.60
Chocolates & Candies	11.60
Home & Cleaning	9.79
Fruits & Vegetables	6.45
Personal Care	6.12
Paan Corner	6.12

Table 3: SQL Result — Stockout Percentage by Category (sorted by severity)

Critical findings:

- Biscuits has the highest stockout rate at 28.57% — nearly 1 in 3 SKUs unavailable, a critical gap for a high-frequency purchase category.
- Dairy, Bread & Batter and Beverages both sit at 21.71% — staple daily-use categories where 1-in-5 unavailability severely undermines the platform's promise.
- Meats, Fish & Eggs is at 19.05%, close to the high-risk threshold despite a small catalog — suggesting supplier or cold-chain reliability issues.
- Cooking Essentials (12.45%) and Munchies (12.45%) are both above the platform average of 12.14%, in medium-risk territory.
- Paan Corner and Personal Care are the best-performing categories at 6.12% — well below the platform average.

A key compounding risk: Dairy, Bread & Batter and Beverages are mid-sized categories (3.46% catalog share each) but carry very high stockout rates. Customers seeking these essentials face a 1-in-5 chance of unavailability — a significant trust and retention risk.

7. Recommendations

- Implement category-specific safety stock thresholds, especially for High and Medium risk categories identified in this analysis.
- Set up automated alerts when a category's out-of-stock rate crosses 10% (Medium) and 20% (High), triggering restocking workflows.
- Align discount and promotional campaigns with inventory availability forecasts to prevent demand-driven stockouts.
- Conduct deeper time-series analysis (if timestamped data is available) to identify whether stockouts follow weekly/daily patterns.

— End of Report —